IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-065

1. Topic (12 words max)

Multi-Modal Machine Learning Approach for Threat Detection in WhatsApp and Telegram: Spam Links, Document Based Malware, Chat Behavior, and Sensitive Content Analysis

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

During the COVID-19 pandemic, many school children in Sri Lanka under the age of 16 started using messaging apps like WhatsApp and Telegram for online learning and communication. However, this sudden increase in online activity also exposed them to new risks such as receiving spam or harmful links, downloading malicious documents, getting involved in suspicious or harmful chats, and being exposed to sensitive images and videos.

Most of these children are not aware of how to stay safe online. They can easily click on a fake link, download a harmful file, or be emotionally affected by harmful conversations or sensitive media. Since parents and teachers may not always be able to monitor these apps, there is a growing need for a smart, easy-to-use solution that can automatically detect and alert users about these threats. That's why we propose to implement this mobile app.

References :

[1] *Youtu.be*, 2025. https://youtu.be/jsBzfQvfpI4?si=VrwWxgn3wxCHmT1Q
[2] "Facebook," *Facebook.com*, 2022. https://www.facebook.com/share/14zyUouURY/ (accessed Jun. 26, 2025).
[3] "Facebook," *Facebook.com*, 2022. https://www.facebook.com/share/1AdJy77DpP/ (accessed Jun. 26, 2025).
[4] "Facebook," *Facebook.com*, 2022. https://www.facebook.com/share/19H7fwNUuP/ (accessed Jun. 26, 2025).
[5] "Facebook," *Facebook.com*, 2019. https://www.facebook.com/share/16iFZAqnpg/ (accessed Jun. 26, 2025).
[6] "Facebook," *Facebook.com*, 2022. https://www.facebook.com/share/1EiarVcGDE/ (accessed Jun. 26, 2025).

[7] "Facebook," *Facebook.com*, 2022. https://www.facebook.com/share/1AUhvoedXX/ (accessed Jun. 26, 2025).

[8] "Sri Lanka CERT|CC," *Facebook.com*, 2022. https://www.facebook.com/share/p/1GxNdEEmWC/ (accessed Jun. 26, 2025).

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

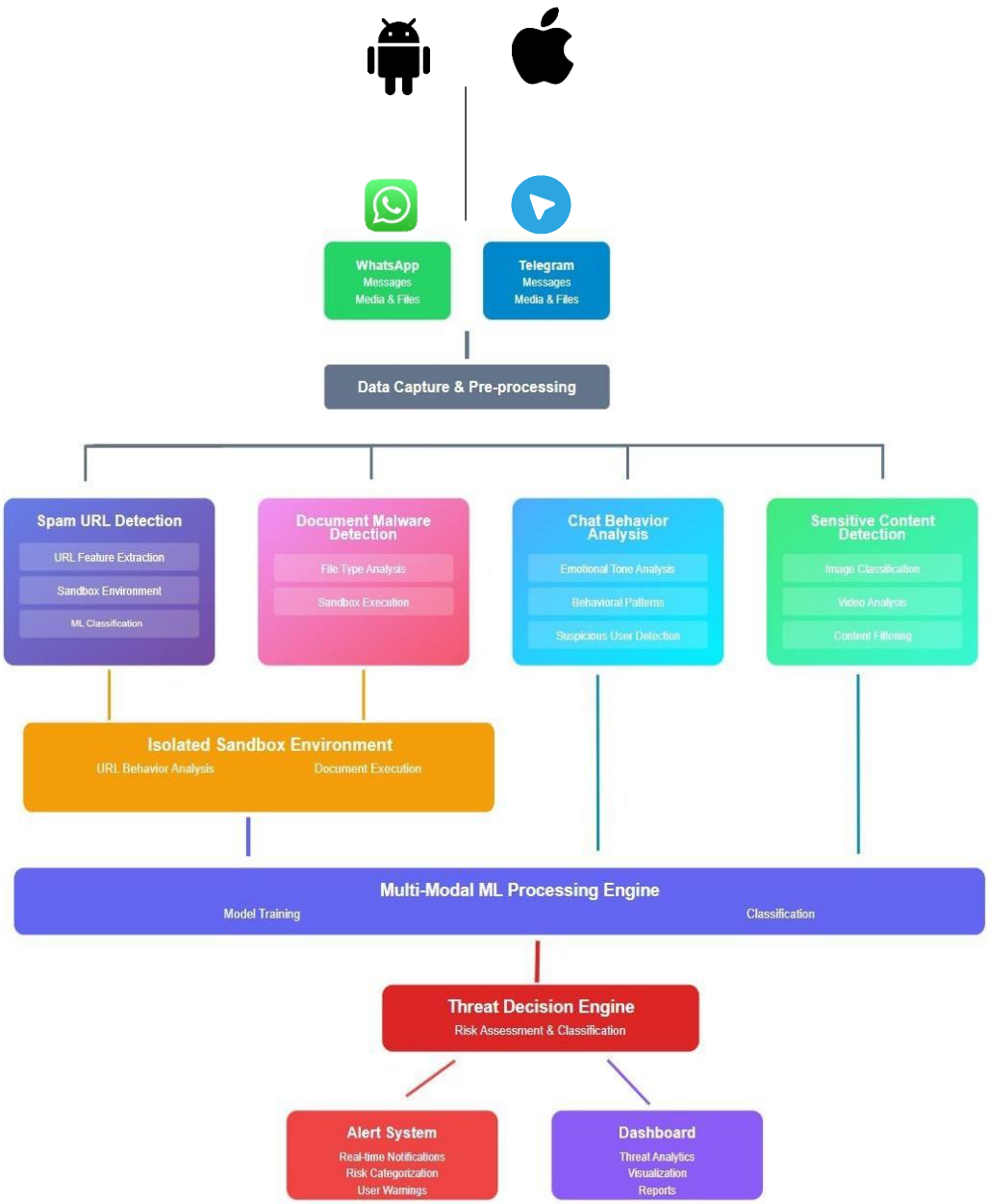
This project proposes the development of a **cross-platform mobile application** for both **Android and iOS**, designed to detect and prevent cybersecurity threats in **WhatsApp and Telegram** using **multi-modal machine learning** and **sandbox-based analysis**. The system aims to be **lightweight and resource-efficient**, ensuring smooth performance on a wide range of mobile devices.

The application is composed of **four main detection modules**:

1. **Spam & Malicious Link Detector** – Uses a sandbox environment to simulate and classify URL behavior from shared messages.
2. **Document-Based Malware Detector** – Analyzes PDFs, DOCs, and APKs in a sandbox to identify threats based on runtime behavior.
3. **Chat Behavior & Emotion Analyzer** – Applies NLP techniques and behavioral profiling to detect fake or suspicious user interactions.
4. **Sensitive Media Filter** – Uses computer vision models to detect nudity, violence, or explicit content in outgoing images/videos and sends alerts to the sender.

The system processes incoming messages and media, extracts relevant features, and uses pre-trained machine learning models optimized for **real-time, low-latency operation**. Users receive visual alerts and warnings through a simple and user-friendly interface.

Conceptual Diagram



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project requires multidisciplinary expertise across **Cybersecurity**, **Machine Learning (ML)**, **Natural Language Processing (NLP)**, and **Computer Vision**, along with an understanding of **child online safety in the Sri Lankan context**. Each component demands focused technical and domain-specific knowledge to address growing digital threats in messaging apps used by minors.

- **Spam & URL Detection**: Requires knowledge of phishing patterns, sandbox environments, URL behavior analysis, and malicious link classification using dynamic analysis tools.
- **Document-Based Malware Detection**: Involves malware analysis using sandboxing, behavior-based threat detection, and feature engineering from file execution logs (for PDFs, DOCs, and APKs). Understanding macro viruses and embedded script behavior is essential.
- **Emotion & Behavioral Analysis**: Requires skills in emotion detection using NLP, chat activity profiling, and anomaly detection to identify fake or manipulative accounts based on conversation tone and timing patterns.
- **Sensitive Content Detection**: Demands expertise in computer vision, deep learning, and mobile-optimized image/video classification techniques to detect nudity, violence, and explicit content in real-time on encrypted messaging platforms.

In terms of **data requirements**, the project will use:

- Labeled chat datasets (spam, emotion, behavioral patterns)
- Malware-infected file datasets and sandbox outputs
- Image/video datasets for sensitive content detection
- Phishing URL datasets and simulated WhatsApp/Telegram traffic

Public datasets from **Kaggle, UCI, OpenML**, and sandbox services (like **Any.Run, Cuckoo Sandbox**) will be used. **Synthetic datasets** will be ethically generated where real data is sensitive or unavailable.

This research aligns with national cybersecurity goals. Guidelines from **Sri Lanka CERT|CC**, **Ministry of Women and Child Affairs**, and the **CID Cyber Crime Division** will inform the social and legal aspects of the system. We will follow strict ethical standards in simulating data and designing privacy-aware features, especially given our focus on protecting minors.

8. Objectives and Novelty

Main Objective

To develop a mobile application that leverages multi-modal machine learning techniques and sandbox-based analysis to detect various threats in WhatsApp and Telegram, including spam links, document-based malware, suspicious chat behavior, and sensitive multimedia content, in order to enhance user safety and prevent misuse of messaging platforms.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Gangoda G W A P – IT22594440	Detection and Classification of Spam URL-Based Messages in WhatsApp and Telegram Using a Sandbox Environment	1. Collect and label WhatsApp and Telegram messages containing URLs by simulating chat data and referencing known threat sources. 2. Design and implement a sandbox environment to safely analyze the behavior of embedded URLs in real-time. 3. Extract and engineer features from both message content and URL behaviors observed in the sandbox. 4. Extract features from both message content and URL behaviors observed in the sandbox. 5. Integrate the detection system with a user interface that provides classification results alert.	This project introduces a custom sandbox environment to detect malicious URLs in WhatsApp and Telegram messages. Instead of just analyzing message text, the system safely opens and observes URL behavior to catch hidden threats. This dynamic method improves detection accuracy and offers a more secure way to identify dangerous links shared on WhatsApp and Telegram.

Hitihaminelage K I S – IT22561220	Detection of Malware-Infected Documents in Messaging Apps Using Sandbox-Based Behavior Analysis	1. Collect document files (PDFs, DOCs, and APKs) shared through WhatsApp and Telegram chats. 2. Execute each document in a sandbox environment to simulate real-world behavior. 3. Extract behavioral features such as file modifications, API calls, registry changes, and network activity. 4. Label documents as malicious or benign based on known threat behavior or external threat intelligence sources. 5. Train separate machine learning models for each file type or a unified model with type-aware features. 6. Evaluate the accuracy, precision, and recall of the model on each document category. 7. Integrate the detection system into the messaging workflow to identify and flag suspicious file-sharing activity.	This component introduces a novel approach by integrating sandbox-based behavioral analysis into messaging platforms (WhatsApp and Telegram) to detect malware embedded in shared documents. Unlike traditional static antivirus scans, this system simulates real document execution behavior in a controlled environment to uncover hidden threats like macro viruses, embedded scripts, and polymorphic malware a feature rarely addressed in chat-based malware detection.
Rupasinghe P S N A – IT22560230	Study user behavior and emotions in Telegram and WhatsApp messages to detect fake accounts.	1. Analyze emotional tone in messages using ML techniques. 2. Extract behavioral features like message timing, frequency, and repetition.	Uniquely compares the emotional and behavioral patterns of real versus fake and suspicious users, providing new insights into how deceptive accounts differ in

		3. Combine emotional and behavioral patterns to identify suspicious users. 4. Train and evaluate ML models for detecting fake or harmful Telegram and whatsapp profiles. 5. Visualize and interpret results through dashboards.	message tone and activity, an area that has been largely unexplored in Telegram and whatsapp-specific research.
Perera N T K D – IT22602428	ML-Based Sensitive Content Detection and Alert System for WhatsApp and Telegram.	1. Collect and curate a dataset of images and videos containing sensitive and non-sensitive content (e.g., nudity, violence, explicit scenes) for training and testing. 2. Develop and train a machine learning model to accurately classify sensitive content using image/video classification techniques. 3. Design a real-time detection system that can scan outgoing media from WhatsApp and Telegram before it is sent. 4. Implement a risk-based alert mechanism that provides users with tiered feedback (e.g., warning, soft block, allow) based on model confidence levels. 5. Optimize the model for lightweight performance, enabling efficient use	An Open-Source ML Framework for Sensitive Content Detection and Alerting in Messaging Apps - While existing solutions for detecting sensitive content on messaging platforms are typically commercial, paid, or closed-source, this component introduces a free and open-source alternative that leverages machine learning to detect nudity and other sensitive content in images and videos shared via WhatsApp and Telegram.

		on mobile devices without compromising user experience. 6. Release the tool as an open-source framework to allow further community-driven improvement, transparency, and accessibility. 7. Ensure ethical handling of user data and privacy, especially while analyzing sensitive content in encrypted platforms.	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Gangoda G W A P – IT22594440	Focuses on cybersecurity in messaging platforms by detecting spam and malicious URLs in WhatsApp and Telegram using sandbox-based analysis. This aligns with secure communication practices and URL-based threat mitigation—key areas in cyber threat intelligence and phishing protection.
Hitihaminelage K I S – IT22561220	Applies malware analysis and cybersecurity principles to detect infected documents (PDFs, DOCs, APKs) shared in WhatsApp and Telegram. Uses sandbox environments and behavioral analysis to uncover hidden threats, contributing to secure file handling and content filtering mechanisms.
Rupasinghe P S N A – IT22560230	Enhances platform safety by using machine learning and behavioral analytics to identify fake or suspicious users through emotional tone and chat behavior analysis, supporting early detection of social engineering and manipulation—a growing concern in cybersecurity.
Perera N T K D – IT22602428	Implements sensitive content detection for images and videos in messaging apps using deep learning. Provides real-time alerts for inappropriate media, helping prevent digital exploitation and aligning with ethical cybersecurity practices, especially in child-safe communication systems.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Deemantha	Siriwardana	
Co-Supervisor	Miss.	Ayesha	Wijesooriya	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July

Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.